Deploying Java Web Applications on CloudSwing Open PaaS

Development and deployment are slow in traditional environments due to issues with team synchronisation across regions, software version and licensing management (for proprietary products), backup and disaster-recovery management, and access to the required computing, storage and network resources. Cloud computing provides various ways to address these challenges; one of these is PaaS (Platform-as-a-Service). This article explains CloudSwing, a solution that "lets you deploy any application on any technology stack to any cloud."

PaaS is application middleware (Web servers, application servers and databases) offered as a service—which would otherwise have to be added to the basic infrastructure in a virtual machine on the cloud. Thus, PaaS provides a platform for the quick and flexible deployment and management of business applications, with standardised middleware services. This is done by leveraging standard, pre-built technology stacks. Leading organisations are already using cloud PaaS services.

The PaaS market is growing fast. In 2011, Gartner's special report: 'PaaS Road Map: A Continent Emerging' included the following prediction, "All major software vendors—including IBM, Oracle, Microsoft, SAP, Red Hat, salesforce.com, Google and VMware—will deliver new and notable technology in this market, making 2011 the 'Year of PaaS'. However, most offerings will require additional investment to reach maturity." At the start of 2011, most vendors active in the cloud application platform as a service (aPaaS) market were cloud specialists (e.g., salesforce.com and Google). By the end of 2011, the market had changed dramatically—IBM, Oracle, Red Hat, SAP and VMware announced plans for aPaaS offerings.

Java PaaS offerings

Java PaaS has matured in the past few months with product offerings evolving rapidly. Google App Engine was a solitary Java PaaS provider. Most PaaS offerings were for platforms like Ruby and Python, but fortunately, that has started to change. In the past year or so, quite a few commercial providers have entered the Java PaaS space—CloudSwing, CloudBees, Amazon Elastic Beanstalk, Cloud Foundry, Heroku, and Red Hat OpenShift. This makes sense, since Java developers are one of the biggest developer groups in the world.

The Java platform is well-suited for PaaS, since the JVM, application server, and deployment archives (e.g., WARs and EARs) provide isolation for Java applications, allowing several developers to deploy applications on the same infrastructure.

CloudSwing: Deploying to the cloud within minutes

Conventional PaaS offerings limit the user's choice to the service provider's offerings. Open PaaS eliminates this lock-in by letting users customise technology stacks, components and versions, and choose cloud service providers or deployment models to optimise performance, scalability, security and costs for each application. CloudSwing, from